Experiment-17 : Eye diagram analysis of optical communication link using optisystem

AIM:

To design and simulate an optical communication link using OptiSystem and analyze the eye diagram to evaluate the signal quality and transmission performance.

OBJECTIVES:

- To design an optical communication system using OptiSystem.
- To generate and transmit an optical signal through optical fiber.
- To observe the received signal using an eye diagram.
- To analyze signal quality, noise and inter-symbol interference (ISI).
- To study the performance of the optical link.

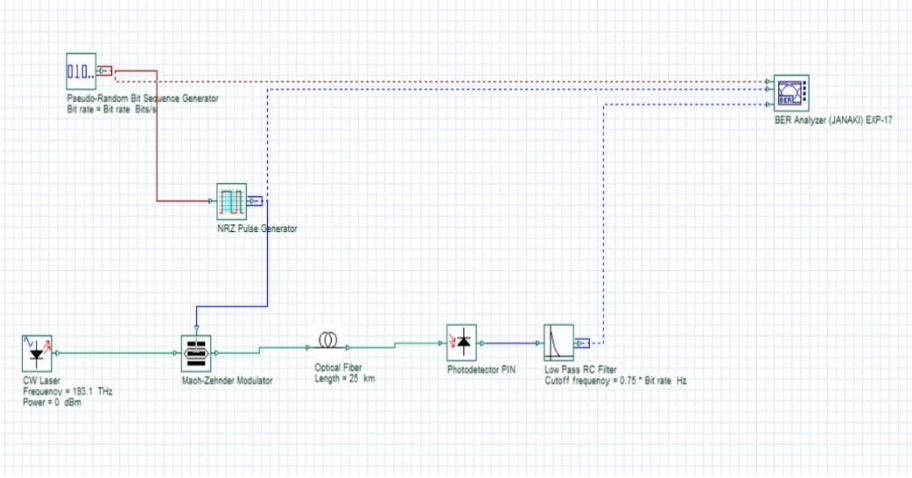
SOFTWARE REQUIRED:

- Optisystem

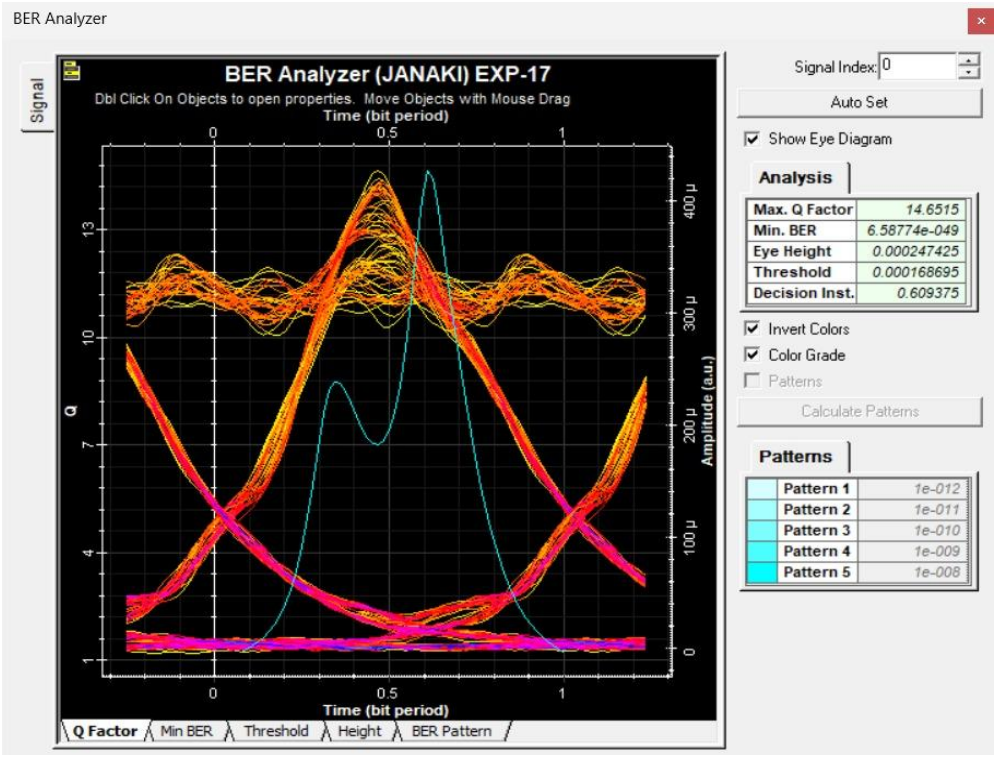
EXPERIMENT PROCEDURE:

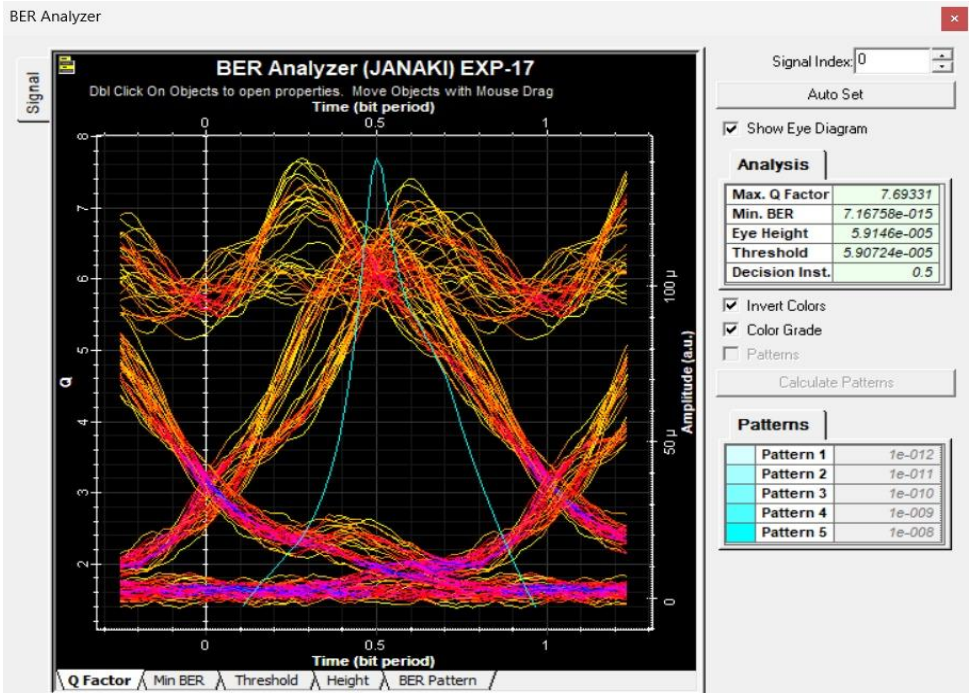
- Open the OptiSystem software and create a new project.
- Select the required components from the Component Library.
- Place the PRBS Generator and NRZ Pulse Generator.
- Connect the optical transmitter (Laser) and Mach-Zehnder Modulator.
- Connect the optical fiber to transmit the signal.
- Connect the PIN Photodetector and Low Pass Filter at the receiver section.
- Add the BER Analyzer to observe the eye diagram.
- Connect all components according to the optical communication link.
- Run the simulation by clicking the Calculate button.
- Open the BER Analyzer and select Show Eye Diagram.
- Observe the eye opening, Q-factor and BER values.
- Record the results and analyze the signal quality.

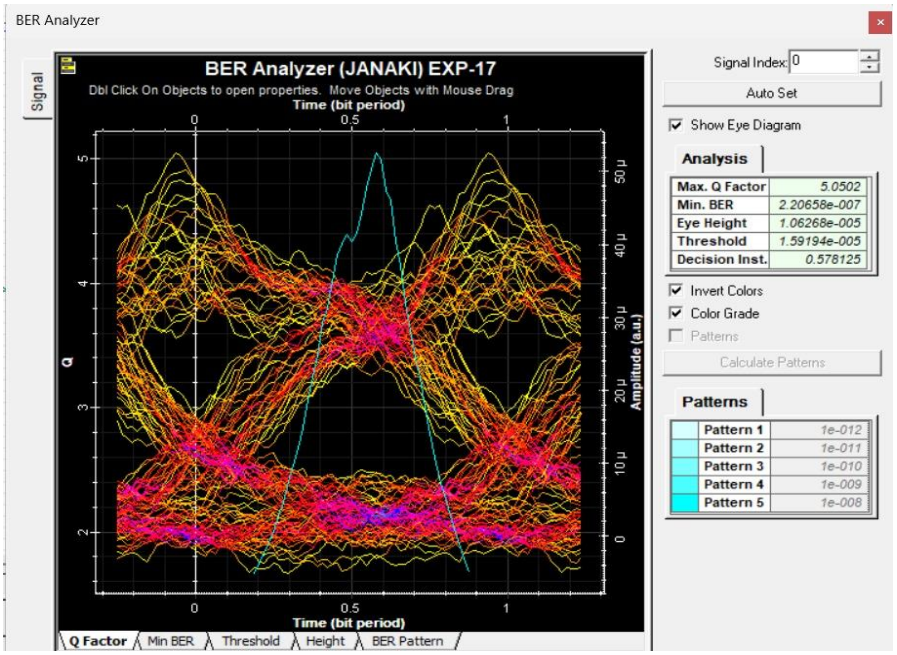
BLOCK DIAGRAM:

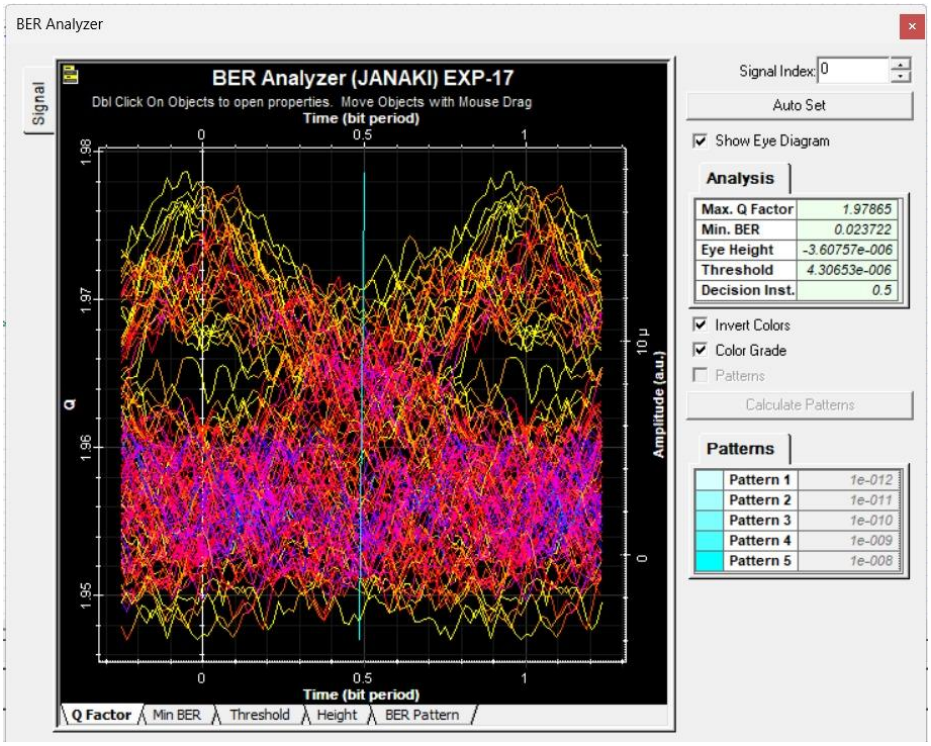


OUTPUT WAVEFORMS:






BER Analyzer
BER Analyzer (JANAKI) EXP-17
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 5.0502
Min. BER 2.20658e-007
Eye Height 1.06268e-005
Threshold 1.59194e-005
Decision Inst. 0.578125
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern


BER Analyzer
BER Analyzer (JANAKI) EXP-17
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Signal
Q
Amplitude (a.u.)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 1.97865
Min. BER 0.023722
Eye Height -3.60757e-006
Threshold 4.30653e-006
Decision Inst. 0.5
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q Factor
Min BER
Threshold
Height
BER Pattern

REFLECTION:

The experiment helped to understand the performance of an optical communication link using OptiSystem. The eye diagram was used to observe signal quality, noise and inter-symbol interference.

RUBRICS:

The experiment helped to understand the performance of an optical communication link using OptiSystem. The eye diagram was used to observe signal quality, noise and inter-symbol interference

Assessment Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Weightage
System Design	Correct optical link design in OptiSystem.	Minor connection errors.	Partial system setup.	Incorrect setup.	20%
Parameter Configuration	All transmitter, fiber and receiver parameters correctly configured.	Minor parameter errors.	Some parameters missing.	Incorrect configuration.	20%
Measurement & Monitoring	Accurate eye diagram, BER and Q-factor readings.	Minor missing readings.	Partial measurements.	No proper measurements.	20%
Signal Quality Analysis	Correct analysis of eye opening, noise and ISI.	Basic signal analysis.	Limited analysis.	No proper analysis.	20%
Report & Conclusion	Complete report with eye diagram, graphs and results.	Good report and conclusion.	Incomplete report.	Poor documentation.	20%

RESULT:

The optical communication link was successfully designed and simulated using OptiSystem. The eye diagram was observed and analyzed to evaluate the signal quality and transmission performance.